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Area, Pythagoras and similar shapes

Level 2 Intermediate · Geometry

How to find lengths and areas by subtraction, Pythagoras and its triples, and the ratios of similar triangles.

By the end you can: find awkward areas by subtraction and by comparing triangles that share a height; recognise Pythagorean triples and set up Pythagoras equations, including in folding problems; spot similar triangles and use their length and area ratios.

Lesson

How far is corner B of a 6 by 8 rectangle $ABCD$ from the diagonal AC? You could set up coordinates and find the equation of a line, but there is a one-line answer. The diagonal has length 10, because 6, 8, 10 is a right-angled triangle. Triangle ABC has area $\frac{1}{2}\times 6\times 8 = 24$. Its area is also $\frac{1}{2}\times AC\times h = 5h$, where h is the distance we want. So $h = \frac{24}{5}$.

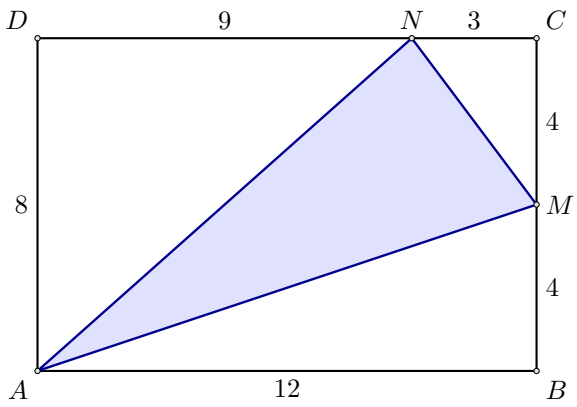
Computing one area in two ways, recognising a Pythagorean triple, and spotting similar shapes: these are the three tools of this unit, and between them they handle a large share of the geometry on the IMC, the Kangaroos and the AMC 10.

Area: decompose, subtract, compare

The area of a triangle is $\frac{1}{2}\times$ base $\times$ height, where the height is measured perpendicular to the base, and may fall outside the triangle. Three consequences do most of the work.

> **Area facts.** (1) Triangles on the same base with the same height have equal areas, so sliding the apex parallel to the base does not change the area. (2) Triangles with the same height have areas in the ratio of their bases. (3) A triangle has half the area of a parallelogram on the same base and between the same parallels.

When a region is awkward, find it by subtracting from something simple, such as the rectangle around it, or by cutting it into pieces whose bases and heights you know. Fact (2) is the one people forget: if D lies on BC with $BD : DC = 2 : 3$, then triangles ABD and ADC have areas in the ratio $2 : 3$, whatever the shape of the triangle, because they share the height from A.



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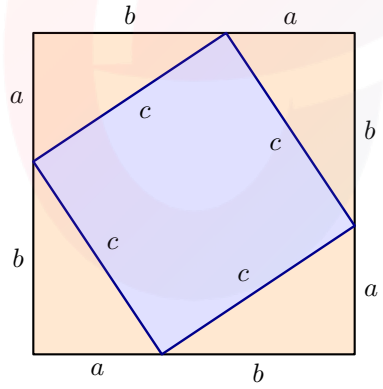
Example 1. Rectangle $ABCD$ has $AB = 12$ and $BC = 8$. M is the midpoint of BC, and N lies on CD with $CN = 3$. Find the area of triangle AMN.

Solution.
Using $\frac{1}{2} \times$ base $\times$ height directly needs the distance from N to the line AM, which is unpleasant to find. But the triangle sits inside the rectangle, and the three pieces left over are right-angled triangles with their legs along the sides. Their areas are easy, so subtract.
Triangle ABM has legs 12 and 4, so area 24. Triangle MCN has legs 4 and 3, so area 6. Triangle ADN has legs 8 and 9, so area 36. The rectangle has area 96, so triangle AMN has area $96 - 24 - 6 - 36 = 30$.
This box and subtract method works for any triangle whose vertices lie on the sides of a rectangle, and for any triangle drawn on a square grid: draw the smallest rectangle round it and remove the right-angled triangles (and any rectangles) in the corners.

Pythagoras and the triples worth knowing

Pythagoras' theorem. In a right-angled triangle with legs a, b and hypotenuse c, we have $a^2 + b^2 = c^2$. Conversely, if $a^2 + b^2 = c^2$ then the angle opposite c is a right angle.

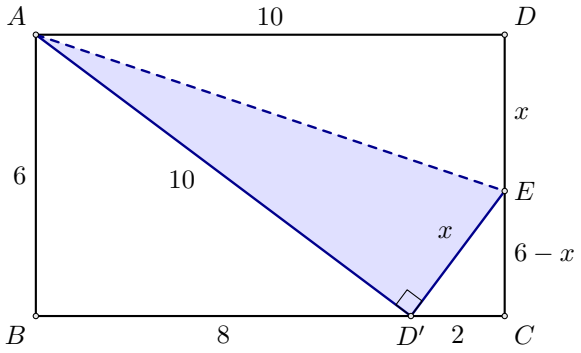
Here is a proof by area. Place four copies of the triangle inside a square of side $a + b$, with their right angles in the corners. At each point where two triangles meet on a side of the big square, they contribute the two different acute angles of the triangle, which add to $90°$, so the angle of the hole there is $180° - 90° = 90°$. The four sides of the hole are hypotenuses, so the hole is a square of side c. Counting the area two ways, $(a + b)^2 = c^2 + 4 \times \frac{1}{2}ab$, which simplifies to $a^2 + b^2 = c^2$.



Whole-number solutions are called Pythagorean triples. Know these on sight: $(3, 4, 5)$, $(5, 12, 13)$, $(8, 15, 17)$, $(7, 24, 25)$, $(20, 21, 29)$ and $(9, 40, 41)$, together with multiples such as $(6, 8, 10)$ and $(15, 20, 25)$. You can manufacture more from $(m^2 - n^2,\ 2mn,\ m^2 + n^2)$, since $(m^2 - n^2)^2 + (2mn)^2 = (m^2 + n^2)^2$; in fact every triple with no common factor arises this way, though that takes more work to prove.

Two right-angled triangles with angles you know are worth memorising. Half a square has sides $1, 1, \sqrt{2}$ and angles $45°, 45°, 90°$. Half an equilateral triangle has sides $1, \sqrt{3}, 2$ and angles $30°, 60°, 90°$. Finally, one algebraic move: $c^2 - a^2 = b^2$ factorises as $(c - a)(c + a) = b^2$, which is the way to find whole-number solutions with one side fixed.

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Example 2. Rectangle $ABCD$ has $AB = 6$ and $BC = 10$. It is folded along a line AE, with E on CD, so that D lands on a point D' of side BC. Find DE.

Solution.
A fold is a reflection in the crease, so lengths and angles are preserved: $AD' = AD = 10$, $ED' = ED$, and $\angle AD'E = \angle ADE = 90°$. Start with what is fully determined. In the right-angled triangle ABD' we know $AB = 6$ and the hypotenuse $AD' = 10$, so $BD' = 8$ (a $6, 8, 10$ triangle) and $D'C = 2$.
Now let $DE = x$. Then $ED' = x$ and $EC = 6 - x$, and Pythagoras in triangle ECD' gives $(6 - x)^2 + 2^2 = x^2$. Expanding, $40 - 12x = 0$, so $x = \frac{10}{3}$.
There is also a route with no algebra. Since $\angle AD'E = 90°$, the angles $\angle AD'B$ and $\angle ED'C$ add to $90°$, so $\angle ED'C = \angle BAD'$. Triangles ABD' and $D'CE$ therefore have the same angles, and the ratio of corresponding sides is $D'C : AB = 2 : 6 = 1 : 3$. So $ED' = \frac{1}{3}AD' = \frac{10}{3}$. Right angles that meet on a straight line almost always produce similar triangles like this.

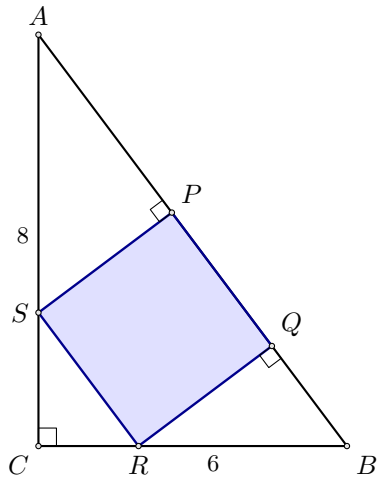
Similar shapes

Two triangles are similar if their angles are equal in pairs; then all corresponding lengths are in the same ratio. For triangles, two pairs of equal angles are enough, since the third pair follows. Write the vertices in matching order: if $\triangle ABC$ is similar to $\triangle PQR$, with $\angle A = \angle P$ and so on, then $\dfrac{AB}{PQ} = \dfrac{BC}{QR} = \dfrac{CA}{RP}$.

Scale factors. If lengths are multiplied by k, areas are multiplied by k^2 and volumes by k^3.

For a triangle this is clear, since the base and the height are both multiplied by k, and any polygon is made of triangles. Three configurations come up again and again. A line parallel to one side of a triangle cuts off a smaller triangle similar to the whole (the A shape). Two parallel segments with their ends joined crosswise give two similar triangles meeting at a point (the X shape), which is what the diagonals of a trapezium do. And the altitude to the hypotenuse of a right-angled triangle splits it into two triangles similar to the whole; if the altitude from C meets AB at D, this gives $CD^2 = AD \times DB$.

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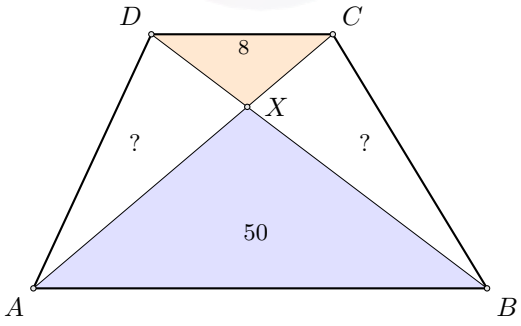
Example 3. Triangle ABC has a right angle at C, with $CA = 8$ and $CB = 6$. A square $PQRS$ has P and Q on AB, R on CB and S on CA. Find the side of the square.

Solution.

First, $AB = 10$. Coordinates would work, but the square is tilted and the equations get messy. Look instead at the three small triangles round the square. Each is right-angled and shares an angle with ABC, so each is similar to ABC. That turns the problem into ratios.

Let the side be s. Triangle APS has a right angle at P and the angle A of the big triangle, so $\dfrac{AP}{PS} = \dfrac{AC}{CB} = \dfrac{8}{6}$, giving $AP = \frac{4}{3}s$. Triangle BQR has a right angle at Q and the angle B, so $\dfrac{BQ}{QR} = \dfrac{BC}{CA} = \dfrac{6}{8}$, giving $BQ = \frac{3}{4}s$. Along the hypotenuse, $\frac{4}{3}s + s + \frac{3}{4}s = 10$, so $\frac{37}{12}s = 10$ and $s = \frac{120}{37}$.

As a check, use the A shape. The altitude from C to AB is $\frac{6\times 8}{10} = \frac{24}{5}$ (area two ways again). Triangle CSR is similar to CAB because SR is parallel to AB, and its height is $\frac{24}{5} - s$, so $\dfrac{s}{10} = \dfrac{24/5 - s}{24/5}$. This also gives $s = \frac{120}{37}$. For comparison, the square standing in the corner at C has side $\frac{24}{7}$, which is larger: a pleasant surprise worth remembering.



Example 4. $ABCD$ is a trapezium with AB parallel to DC. Its diagonals meet at X. Triangle AXB has area 50 and triangle CXD has area 8. Find the area of the trapezium.

Solution.

No lengths are given, and trying to find AB, CD and the height is hopeless: infinitely many trapezia fit the data. So the answer must come from ratios alone.

Since AB is parallel to DC, alternate angles give $\angle XAB = \angle XCD$ and $\angle XBA = \angle XDC$, so

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triangles AXB and CXD are similar (the X shape). Their areas are in the ratio $50 : 8 = 25 : 4$, so their lengths are in the ratio $5 : 2$. In particular $AX : XC = 5 : 2$.
Now use a shared height. Triangles AXD and CXD have their bases AX and XC on the same line and share the apex D, so their areas are in the ratio $5 : 2$, which makes the area of AXD equal to 20. Similarly triangle BXC has area 20. (Alternatively, triangles ABD and ABC have the same base and the same height, so removing AXB from each shows that AXD and BXC have equal areas.) The trapezium has area $50 + 8 + 20 + 20 = 98$.
In general, if the two triangles on the parallel sides have areas P and Q, each side triangle has area $\sqrt{PQ}$ and the trapezium has area $(\sqrt{P} + \sqrt{Q})^2$.

Common traps.

- Scaling areas by k instead of k^2. Lengths in the ratio $2 : 3$ give areas in the ratio $4 : 9$.
- Matching the wrong sides in similar triangles. Pair up the equal angles first, then read off the sides opposite them.
- Using Pythagoras without a right angle, or with the wrong side as the hypotenuse: it is always the side opposite the right angle.
- Taking a slanted length as a height. The height must be perpendicular to the base.
- Seeing a triple that is not there. Check that $a^2 + b^2 = c^2$ before you trust it.

How to spot this in a contest

If right angles appear and lengths are asked for, look for Pythagoras and for triples first: setters love numbers that make triples. If an area is asked for and the region is awkward, subtract it from a rectangle, or compare triangles that share a height. If there are parallel lines, there are similar triangles: mark the equal angles and write down the ratio. If the question gives areas and asks for areas, think in ratios: find a length ratio by similarity, square it, and use shared heights for the rest. Folding problems are reflections in disguise, and almost always lead to a Pythagoras equation in one unknown. And whenever you need the distance from a point to a line, compute an area two ways.

Practice

Try each problem before turning to the hints and solutions at the back.

1. ⋆ A right-angled triangle has hypotenuse 25 and one leg 7. Find its area.

2. ⋆⋆ P is a point inside rectangle $ABCD$. Triangles PAB, PBC and PCD have areas 13, 9 and 20. Find the area of triangle PDA.

3. ⋆⋆ Square $ABCD$ has side 6. E is the point on BC with $BE = 2$, and AE meets the diagonal BD at F. Find the area of triangle BEF.

4. ⋆⋆⋆ How many right-angled triangles with whole-number side lengths have a leg of length 20? (Triangles that are congruent count once.)

5. ⋆⋆⋆ Rectangle $ABCD$ has $AB = 15$ and $BC = 8$. Points P on AB and Q on CD are chosen so that $APCQ$ is a rhombus. Find the length PQ.

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6. ⋆⋆⋆⋆ A right-angled triangle has perimeter 120, and the altitude from the right angle to the hypotenuse has length 24. Find the area of the triangle.

Contest ladder

Real past problems that use this idea, easiest first. Open them on the official paper or the AoPS wiki, or online at kevinvariant08.github.io.

1. AMC 10B 2006 Problem 8 ⋆
2. UKMT IMC 2018 Question 8 ⋆
3. UKMT IMC 2017 Question 13 ⋆⋆
4. AMC 10A 2018 Problem 13 ⋆⋆
5. AMC 10A 2018 Problem 9 ⋆⋆
6. AMC 10A 2008 Problem 20 ⋆⋆⋆
7. UKMT IMC 2018 Question 22 ⋆⋆⋆
8. AMC 10A 2016 Problem 19 ⋆⋆⋆
9. UKMT Maclaurin 2026 Question 2 ⋆⋆⋆
10. AMC 10B 2021 Problem 21 ⋆⋆⋆
11. UKMT Maclaurin 2026 Question 5 ⋆⋆⋆⋆
12. UKMT Hamilton 2026 Question 4 ⋆⋆⋆⋆

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Hints

1. Find the other leg with Pythagoras, or recognise the triple.
2. Add the areas of the two triangles on opposite sides. What do their heights add up to?
3. AD is parallel to BE, so look for an X shape at F.
4. If the other leg is b and the hypotenuse is c, factorise $c^2 - b^2$.
5. Find the side of the rhombus with Pythagoras, then compute the rhombus's area in two ways.
6. Call the legs a, b and the hypotenuse c. The area gives one equation, and squaring $a + b$ brings in Pythagoras.

Solutions

Problem 1

Answer: 84

The other leg is $\sqrt{25^2 - 7^2} = \sqrt{625 - 49} = \sqrt{576} = 24$, so this is the $(7, 24, 25)$ triple. The legs are perpendicular, so the area is $\frac{1}{2} \times 7 \times 24 = 84$.

Problem 2

Answer: 24

Triangles PAB and PCD have bases AB and CD, both equal to the length of the rectangle, and their heights are the distances from P to the opposite sides AB and CD, which add up to BC. So their areas add to $\frac{1}{2} \times AB \times BC$, half the rectangle. The same is true of triangles PBC and PDA. Hence, writing x for the area of PDA, we get $13 + 20 = 9 + x$, so $x = 24$.

(Such a point exists: for example, take $AB = 11$ and $BC = 6$, with P at distance 3 from BC and $\frac{26}{11}$ from AB.)

Problem 3

Answer: $\frac{3}{2}$

Since AD is parallel to BE, triangles FAD and FEB have equal alternate angles and equal vertically opposite angles at F, so they are similar with ratio $AD : EB = 6 : 2 = 3 : 1$. Hence $AF : FE = 3 : 1$, and FE is a quarter of AE.

Triangles BEF and BEA share the apex B, and their bases FE and AE lie on the same line, so the ratio of their areas is $FE : AE = 1 : 4$. Triangle ABE has legs 6 and 2 and area 6, so triangle BEF has area $\frac{6}{4} = \frac{3}{2}$.

Problem 4

Answer: 4

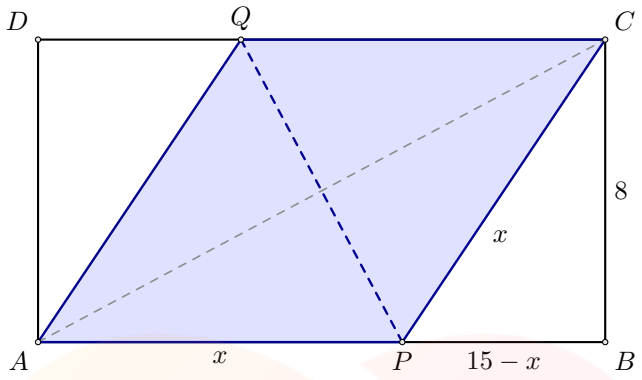
Let the other leg be b and the hypotenuse c. Then $c^2 - b^2 = 400$, that is $(c - b)(c + b) = 400$. The two factors have the same parity (they differ by $2b$), and their product is even, so both are even. Write $c - b = 2u$ and $c + b = 2v$ with $u < v$; then $uv = 100$, and each such factor pair gives $b = v - u$ and $c = u + v$.

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The pairs with $u < v$ are $(1, 100)$, $(2, 50)$, $(4, 25)$ and $(5, 20)$; the pair $(10, 10)$ would give $b = 0$. They give the triangles $(20, 99, 101)$, $(20, 48, 52)$, $(20, 21, 29)$ and $(15, 20, 25)$, so there are 4.

Problem 5

Answer: $\frac{136}{15}$



Let the side of the rhombus be x, so $AP = PC = x$ and $PB = 15 - x$. Pythagoras in triangle PBC gives $(15 - x)^2 + 8^2 = x^2$, so $289 - 30x = 0$ and $x = \frac{289}{30}$.

The rhombus is a parallelogram with base AP and height BC, so its area is $\frac{289}{30} \times 8 = \frac{1156}{15}$. Its diagonals are perpendicular, so its area is also $\frac{1}{2} \times AC \times PQ$, and $AC = 17$ (an $8, 15, 17$ triangle). Hence $PQ = \dfrac{2 \times 1156}{15 \times 17} = \frac{136}{15}$.

Problem 6

Answer: 600

Let the legs be a and b and the hypotenuse c. Computing the area in two ways gives $\frac{1}{2}ab = \frac{1}{2} \times 24c$, so $ab = 24c$. The perimeter gives $a + b = 120 - c$.

Squaring, $a^2 + 2ab + b^2 = (120 - c)^2$. By Pythagoras $a^2 + b^2 = c^2$, so $c^2 + 48c = 14400 - 240c + c^2$, which gives $288c = 14400$ and $c = 50$. The area is $\frac{1}{2} \times 24 \times 50 = 600$. (The legs are 30 and 40: they satisfy $a + b = 70$ and $ab = 1200$.)

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